Worked solutions to

Bayesian Statistical Methods

With Applications to Machine Learning — Reich & Ghosh (2nd ed.)

Chapter 7

Case studies using hierarchical modeling

Every exercise is presented in the same order:

- the **exact** exercise text from the book;
- the problem **in plain words**;
- the **background** you need, recalled from scratch;
- the **key idea** that unlocks the problem;
- a **step-by-step solution** — every algebra step written out, with a *Why?* note whenever a step is not obvious;
- a **sanity check** of the answer;
- the boxed **answer**;
- **takeaways** and common mistakes.

All numbers and figures are reproduced by `ch07/code/ch07.R` (hand-written samplers, base R). The JAGS versions of Exercises 3(b), 4 and 6 are in `ch07/code/ch07_jags.R` and run on GitHub.

Sai Sridhar · October 21, 2026

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Answers at a glance

1	Compatible only if $a = b$ and $\lvert a\rvert < 1$; e.g. $a = 1, b = 0.5$ is incompatible. A DAG multiplies valid conditionals in a fixed order, so the product always integrates to 1.
2	DAG: $(\mu_0, \phi_0, \tau_0) \to Y_t$, $Y_{t-1} \to Y_t$, and $(Y_t, Y_{t-1}, X_{j,t-1}, \alpha_j, \beta_j, \phi_j, \tau_j) \to X_{jt}$.
3	Pooled OR 0.53, random-effects OR 0.43 $(0.19,\ 0.90)$; both say the treatment reduces deaths; RE preferred (WAIC 73.6 vs 78.8).
4	$\rho_i \approx 0.99$ (near random walk); data very informative about θ_1–θ_4; a common-ρ model is simpler and better by WAIC.
5	$\lambda \mid \text{data} \sim \text{Gamma}(a + 165,\ b + 69{,}593)$, ≈ 0.0024/day; men's hazard 1.65 times women's ($\text{Prob} = 0.999$).
6	Range $\rho \approx 66$ km (wide); maps show a high-risk eastern cluster; 10 villages have $\text{Prob}(p_v > 0.5) > 0.9$.

A five-minute primer on the tools of this chapter

Background you need

1. Directed acyclic graphs (DAGs). Each variable is a node; an arrow $A \to B$ means "B's distribution is specified given A". If the graph has no directed cycles, the joint density is the product of each node's density given its parents:

$$p(\text{all nodes}) = \prod_v p(v \mid \text{parents}(v)).$$

Squares will denote observed data, circles unknown parameters, and orange circles hyperparameters.

2. Three layers. *Data layer*: how the observations depend on the process (likelihood). *Process layer*: the scientific model for the unobserved quantities. *Prior layer*: distributions for the parameters.

3. Multivariate random effects. $\boldsymbol{\alpha}_i \overset{\text{iid}}{\sim} \text{N}_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \sim \text{InvWishart}(\nu, \mathbf{S})$ is conjugate: $\boldsymbol{\Sigma} \mid \cdot \sim \text{InvWishart}\big(\nu + K,\ \mathbf{S} + \sum_i (\boldsymbol{\alpha}_i - \boldsymbol{\mu})(\boldsymbol{\alpha}_i - \boldsymbol{\mu})^\top\big)$.

4. Censored data. An observed event at time y contributes the density $f(y)$; a patient still alive at the end of follow-up T contributes the survival probability $\text{Prob}(Y > T)$.

5. Gaussian-process (spatial) random effects. $\boldsymbol{\alpha} \sim \text{N}(\mu\mathbf{1},\ \sigma^2\mathbf{R}(\rho))$ with $R_{uv} = e^{-d_{uv}/\rho}$: nearby locations have similar effects; ρ is the *range* (correlation $e^{-1} \approx 0.37$ at distance ρ). **Elliptical slice sampling** updates the whole vector $\boldsymbol{\alpha}$ without tuning.

Exercise 7.1 — Can we specify a model by its full conditionals?

Exercise 1 (exact text from the book)

Since full conditional distributions are used in many MCMC algorithms, it is tempting to specify the model via its conditional distributions. For example, consider the conditional distributions:

$$Y|X \sim \text{Normal}(aX, 1) \quad \text{and} \quad X|Y \sim \text{Normal}(bY, 1).$$

(a) Select values of a and b so that these full conditional distributions are incompatible; i.e., there is no valid joint distribution for X and Y that gives these full conditional distributions. Argue, but do not formally prove, your assertion that the full conditional distributions are incompatible.

(b) Explain why building a model that produces a valid DAG will always lead to a valid joint distribution.

In plain words

Two conditional distributions do not always fit together into one joint distribution. Find values of a, b where they do not, and explain why building a model top-down (as a DAG) avoids the problem.

Background you need

For a bivariate normal (X, Y): $\text{E}(Y \mid X) = \frac{\text{Cov}(X,Y)}{\text{Var}(X)} X$ and $\text{Var}(Y \mid X) = \text{Var}(Y) - \frac{\text{Cov}(X,Y)^2}{\text{Var}(X)}$ (and symmetrically for $X \mid Y$).

Key idea

If a joint distribution existed, it would be bivariate normal (both conditionals are normal and linear). Matching its slopes and conditional variances to a, b and 1 gives equations that only have a solution for some (a, b).

Solution

(a) Step 1 — what a joint distribution would require.

Let $V_X = \text{Var}(X)$, $V_Y = \text{Var}(Y)$, $C = \text{Cov}(X, Y)$. The conditionals demand

$$\frac{C}{V_X} = a, \qquad \frac{C}{V_Y} = b, \qquad V_Y - \frac{C^2}{V_X} = 1, \qquad V_X - \frac{C^2}{V_Y} = 1.$$

(a) Step 2 — solve.

From the first two, $C = aV_X = bV_Y$. The fourth equation gives $V_X - \frac{a^2 V_X^2}{aV_X/b} = V_X(1 - ab) = 1$, so $V_X = \frac{1}{1-ab}$; by symmetry $V_Y = \frac{1}{1-ab}$ too. Then $aV_X = bV_Y$ forces $\mathbf{a = b}$, and $V_X > 0$ forces $\mathbf{ab < 1}$, i.e. $|a| < 1$.

Why? So the two conditionals are compatible only when $a = b$ with $|a| < 1$ (then X, Y are bivariate normal with variances $\frac{1}{1-a^2}$ and correlation a).

(a) Step 3 — incompatible examples, illustrated by a Gibbs sampler.

- $\mathbf{a = 1,\ b = 0.5}$ $(a \neq b)$. A Gibbs sampler alternating the two conditionals still settles down, but to a distribution whose conditionals are *not* the ones we specified: with 2×10^5 sweeps the output has slope 0.50 for X on Y but slope **0.80** for Y on X (we asked for 1). Theory for the sampler's limit gives exactly $b(1 + a^2)/(1 + b^2) = 0.80$.

- $\mathbf{a = b = 2}$ ($ab > 1$). Each sweep multiplies X by about $ab = 4$: $|X|$ is 380 after 5 sweeps, 4×10^5 after 10 and 4×10^{17} after 30. No distribution can have these conditionals.

Sanity check

Compatible case $a = b = 0.5$: the sampler gives slopes 0.500 and 0.503 and $\mathrm{Var}(X) = 1.327$ vs. the theoretical $1/(1 - ab) = 1.333$. ✓

(b) Why a DAG always works.

Because the graph is acyclic, the nodes can be put in an order in which every node comes after its parents. Each factor $p(v \mid \text{parents}(v))$ is a proper density in v. Integrating the product over the *last* node removes its factor (it integrates to 1, and no other factor depends on it); repeating this for each node in reverse order shows the whole product integrates to 1 — so it is a valid joint density, and its conditionals are exactly the specified ones. A cycle (e.g. $X \to Y \to X$) breaks this argument: there is no "last" node to integrate out first.

Answer

(a) The conditionals are compatible only if $a = b$ and $|a| < 1$. For example $a = 1$, $b = 0.5$ (or $a = b = 2$) is incompatible: matching a bivariate normal's slopes and variances is impossible, and a Gibbs sampler either converges to a distribution with different conditionals (slope 0.80, not 1) or diverges. (b) A DAG defines the joint as a product of proper conditional densities in a parents-first order; integrating out nodes from last to first shows the product integrates to 1.

Takeaway & common mistakes

Specify models top-down (marginal, then conditionals), and let the MCMC algorithm derive the full conditionals — not the other way round.

Exercise 7.2 — A DAG for the climate-reconstruction model

Exercise 2 (exact text from the book)

Draw a DAG for the hierarchical model in Section 7.2.3.

In plain words

Draw the graph of the Brisbane rainfall-reconstruction model: rainfall Y_t follows an AR(1) process, and six proxy records X_{jt} depend on the rainfall in the same year plus their own AR(1) errors.

Background you need

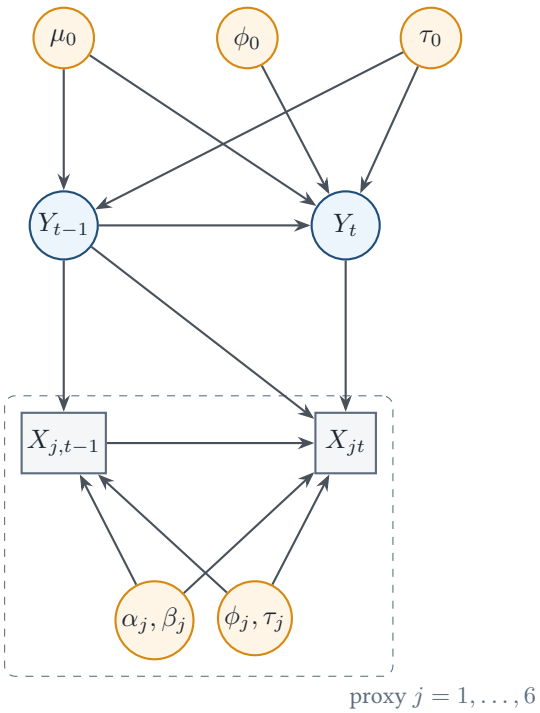
Section 7.2.3 model:

- Rainfall: $Y_1 \sim \mathrm{N}(\mu_0, 1/\tau_0)$, $Y_t \mid Y_{t-1} \sim \mathrm{N}\big(\mu_0 + \phi_0(Y_{t-1} - \mu_0),\ \frac{1-\phi_0^2}{\tau_0}\big)$.
- Proxies: $\mu_{jt} = \alpha_j + \beta_j Y_t$; $X_{j1} \sim \mathrm{N}(\mu_{j1}, 1/\tau_j)$, $X_{jt} \mid X_{j,t-1} \sim \mathrm{N}\big(\mu_{jt} + \phi_j(X_{j,t-1} - \mu_{j,t-1}),\ \frac{1-\phi_j^2}{\tau_j}\big)$.
- Priors on μ_0, ϕ_0, τ_0 and on $\alpha_j, \beta_j, \phi_j, \tau_j$ for $j = 1, \dots, 6$.

Key idea

Draw one arrow into each node from every quantity that appears in its distribution. A *plate* (dashed box) means "repeat for $j = 1, \dots, 6$".

Solution



Reading the graph.

- **Rainfall chain**: Y_t depends on the previous year's rainfall Y_{t-1} and on (μ_0, ϕ_0, τ_0).
- **Proxy chain**: X_{jt} depends on this year's rainfall Y_t (through $\mu_{jt} = \alpha_j + \beta_j Y_t$), on last year's proxy $X_{j,t-1}$ *and* last year's rainfall Y_{t-1} (through $\mu_{j,t-1}$ in the AR term), and on $(\alpha_j, \beta_j, \phi_j, \tau_j)$.

- There are no arrows from proxies to rainfall: rainfall *drives* the proxies. Rainfall values Y_t are partly observed (recent years) and partly missing (they become unknown nodes, imputed by MCMC); the proxies are data (squares), some also missing.

Answer

See the DAG above: $(\mu_0, \phi_0, \tau_0) \rightarrow Y_t \leftarrow Y_{t-1}$; and inside a plate over $j = 1, \ldots, 6$: $(Y_t,\ Y_{t-1},\ X_{j,t-1},\ \alpha_j, \beta_j, \phi_j, \tau_j) \rightarrow X_{jt}$.

Takeaway & common mistakes

Drawing the DAG first makes the code almost write itself: one JAGS line per node, with exactly the parents shown.

Exercise 7.3 — Meta-analysis of corticosteroid trials

Exercise 3 (exact text from the book)

In this problem we will conduct a meta analysis, i.e., an analysis that combines the results of several studies. The data are from the rmeta package in R:

	name	ev.trt	n.trt	ev.ctrl	n.ctrl
1	Auckland	36	532	60	538
2	Block	1	69	5	61
3	Doran	4	81	11	63
4	Gamsu	14	131	20	137
5	Morrison	3	67	7	59
6	Papageorgiou	1	71	7	75
7	Tauesch	8	56	10	71

The data are from seven randomized trials that evaluate the effect of corticosteroid therapy on neonatal death. For trial $i \in \{1, \ldots, 7\}$ denote Y_{i0} as the number of events in the N_{i0} control-group patients and Y_{i1} as the number of events in the N_{i1} treatment-group patients.

(a) Fit the model $Y_{ij}|\theta_j \overset{indep}{\sim} \text{Binomial}(N_{ij}, \theta_j)$ with $\theta_0, \theta_1 \sim \text{Uniform}(0, 1)$. Can we conclude that the treatment reduces the event rate?

(b) Fit the model $Y_{ij}|\theta_{ij} \overset{indep}{\sim} \text{Binomial}(N_{ij}, \theta_{ij})$ with $\text{logit}(\theta_{ij}) = \alpha_{ij}$ and $\boldsymbol{\alpha}_i = (\alpha_{i0}, \alpha_{i1})^T \overset{\text{iid}}{\sim} \text{Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\mu} \sim \text{Normal}(0, 10^2\mathbf{I}_2)$, and $\boldsymbol{\Sigma} \sim \text{InvWishart}(3, \mathbf{I}_2)$. Summarize the evidence that the treatment reduces the death rate.

(c) Discuss the advantages and disadvantages of both models.

(d) Which model is preferred for these data?

In plain words

Seven trials of steroids for premature babies. Combine them (a) by pooling all patients, and (b) with a random-effects model that lets each trial have its own baseline risk and treatment effect.

Background you need

- Beta–binomial (Chapter 1). Odds ratio OR $= \frac{\theta_1/(1-\theta_1)}{\theta_0/(1-\theta_0)}$; on the logit scale, the log odds ratio is $\alpha_{i1} - \alpha_{i0}$.
- Inverse-Wishart update: primer, point 3.

Key idea

Model (a) treats all control patients as one big group and all treated patients as another. Model (b) compares treatment with control *within* each trial and lets trials differ; the overall effect is $\mu_1 - \mu_0$.

Solution

(a) Pooled model.

Totals: control $120/1004$, treatment $67/1007$. Posteriors $\theta_0 \sim \text{Beta}(121,\ 885)$ (mean 0.120) and $\theta_1 \sim \text{Beta}(68,\ 941)$ (mean 0.067).

$$\text{Prob}(\theta_1 < \theta_0 \mid \mathbf{Y}) = 0.99997, \qquad \text{OR: median } 0.53,\ 95\%\ \text{CI } (0.39,\ 0.72).$$

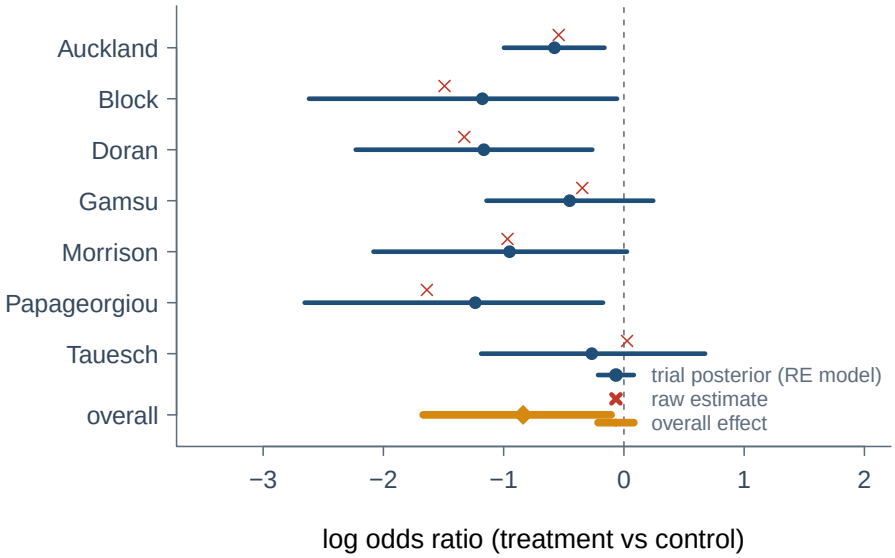
Under this model the evidence is overwhelming: the death rate is roughly halved.

› (b) Random-effects model — sampler.

Each iteration: (i) a Metropolis step for all seven $\boldsymbol{\alpha}_i$ at once (acceptance 47%), using the binomial likelihood and the $\mathrm{N}_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ prior; (ii) $\boldsymbol{\mu} \mid \cdot \sim \mathrm{N}_2\big(\mathbf{V}\boldsymbol{\Sigma}^{-1}\sum_i \boldsymbol{\alpha}_i,\ \mathbf{V}\big)$ with $\mathbf{V} = (7\boldsymbol{\Sigma}^{-1} + \mathbf{I}/100)^{-1}$; (iii) $\boldsymbol{\Sigma} \mid \cdot \sim \mathrm{InvWishart}\big(10,\ \mathbf{I} + \sum_i(\boldsymbol{\alpha}_i - \boldsymbol{\mu})(\boldsymbol{\alpha}_i - \boldsymbol{\mu})^\top\big)$. Three chains × 35,000 kept draws; $\hat{R} = 1.000$ and ESS = 11,227 for $\mu_1 - \mu_0$.

› (b) Results.

$\mu_1 - \mu_0$: mean -0.85, 95% CI $(-1.67,\ -0.11)$, $\quad \mathrm{Prob}(\mu_1 < \mu_0 \mid \mathbf{Y}) = 0.986$, $\quad$ OR median 0.43.



Six of seven trials individually favour treatment; the small trials (Block, Doran, Papageorgiou) are pulled toward the overall effect; Tauesch (the only trial with a positive raw log-OR) is pulled down to -0.26.

Sanity check

JAGS output (3 chains × 40,000 draws).

	Mean	95% interval	$\hat{R}$	ESS
$\mu_1 - \mu_0$ (log OR)	-0.846	$(-1.661,\ -0.097)$	1.001	16,520
μ_0 (control logit)	-2.004	$(-2.49,\ -1.54)$	1.000	20,920
μ_1 (treatment logit)	-2.850	$(-3.61,\ -2.19)$	1.001	16,570

$\mathrm{Prob}(\mu_1 < \mu_0 \mid \mathbf{Y}) = 0.985$ — the same as our sampler $(-0.850,\ 0.986)$.

› (c) Advantages and disadvantages.

- **Pooled model** — *pros*: simple, two parameters, precise. *Cons*: ignores that trials differ (baseline mortality ranges from 8% to 18% across control arms), so it can be biased if treatment/control ratios differ between trials (Simpson's paradox), and its interval is too narrow; it also discards the randomisation *within* trials.
- **Random-effects model** — *pros*: compares arms within each trial (respecting randomisation), allows between-trial heterogeneity in both baseline risk and effect, gives trial-specific shrunken estimates. *Cons*: more complex, needs MCMC, results depend on the prior for $\boldsymbol{\Sigma}$ (only 7 trials to estimate a 2×2 covariance), and the overall effect is less precise.

(d) Which is preferred?

WAIC (over the 14 trial-arm counts): pooled 78.8, random effects **73.6**. The random-effects model fits clearly better, and it is the scientifically appropriate design for a meta-analysis.

Answer

(a) Yes under the pooled model: $\text{Prob}(\theta_1 < \theta_0) \approx 1$, OR 0.53 (0.39–0.72). (b) The random-effects model also supports a benefit: log-OR -0.85 $(-1.67, -0.11)$, $\text{Prob}(\text{reduction}) = 0.986$, OR ≈ 0.43, with a wider interval. (c) Pooled: simple but ignores heterogeneity and within-trial randomisation; RE: realistic but more complex and prior-sensitive. (d) The random-effects model (WAIC 73.6 vs. 78.8).

Takeaway & common mistakes

Meta-analyses should compare arms *within* studies. Pooling raw counts across studies can mislead when studies differ in size and baseline risk.

Exercise 7.4 — A hierarchical model for marathon pacing

Exercise 4 (exact text from the book)

Download the marathon data of Section 6.4 from the course webpage. Let Y_{ij} be the speed of runner i in mile j. Fit the hierarchical model $Y_{i1} \sim \text{Normal}(\mu_i, \sigma_0^2)$ and

$$Y_{ij}|Y_{ij-1} \sim \text{Normal}(\mu_i + \rho_i(Y_{ij-1} - \mu_i), \sigma_i^2),$$

where $\mu_i \overset{\text{iid}}{\sim} \text{Normal}(\theta_1, \theta_2)$, $\rho_i \overset{\text{iid}}{\sim} \text{Normal}(\theta_3, \theta_4)$, and $\sigma_i^2 \overset{\text{iid}}{\sim} \text{InvGamma}(\theta_5, \theta_6)$.

(a) Draw a DAG for this model, and give an interpretation for each parameter in the model.

(b) Select uninformative prior distributions for $\theta_1, \ldots, \theta_6$.

(c) Fit the model in JAGS using three chains each with 25,000 iterations, and thoroughly assess MCMC convergence for the θ_j.

(d) Are the data informative about the θ_j? That is, are the posterior distributions more concentrated than the prior distributions?

(e) In light of (c) and (d), are there any simplifications you might consider, and if so, how would you compare the full and simplified models?

In plain words

Each runner has a typical pace μ_i, a "stickiness" ρ_i (how much last mile's deviation carries over) and a mile-to-mile variability σ_i^2; these vary across runners according to hyperparameters $\theta_1, \ldots, \theta_6$.

Background you need

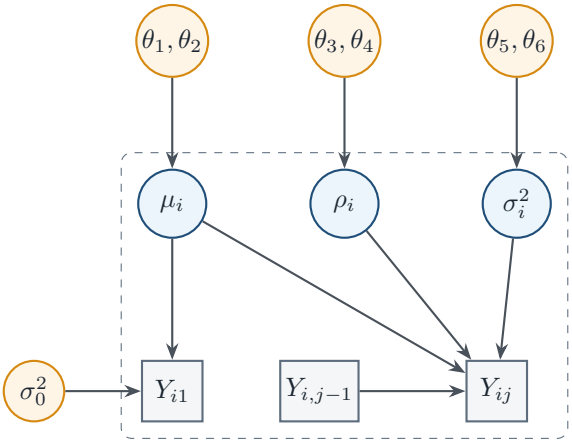
- Data: 2016 Boston Marathon, top female runners, pace in minutes per mile for miles 1–26 (book website). We use the 58 runners with no missing splits (as in Section 2.1.7).
- Given ρ_i, the mean structure is linear in μ_i: $Y_{ij} - \rho_i Y_{i,j-1} = (1-\rho_i)\mu_i + \varepsilon_{ij}$. Given μ_i, it is a regression through the origin of $e_{ij} = Y_{ij} - \mu_i$ on $e_{i,j-1}$ with slope ρ_i. So almost everything is conjugate.

Key idea

Gibbs sampling for μ_i, ρ_i (normal), σ_i^2, σ_0^2 (inverse-gamma), θ_1, θ_3 (normal), θ_2, θ_4 (inverse-gamma), θ_6 (gamma); a Metropolis step for the shape θ_5.

Solution

(a) DAG.



(a) Interpretation.

- μ_i: runner i's long-run average pace (min/mile); ρ_i: persistence — the fraction of a fast/slow deviation that carries into the next mile ($\rho_i \approx 1$: the runner drifts; $\rho_i \approx 0$: each mile bounces back to μ_i); σ_i^2: runner i's mile-to-mile variability; σ_0^2: variability of the first mile around μ_i.
- θ_1, θ_2: mean and variance of the typical pace across runners; θ_3, θ_4: mean and variance of persistence across runners; θ_5, θ_6: shape and scale of the distribution of the runners' variabilities.

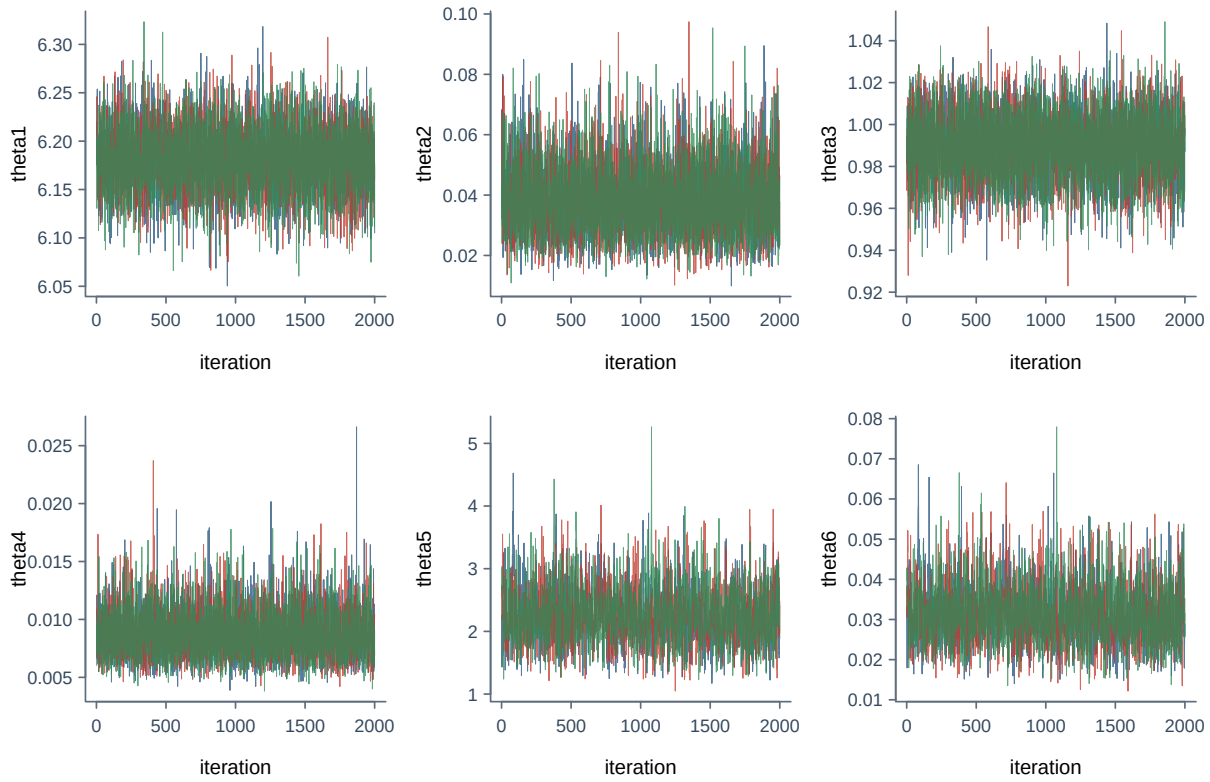
(b) Uninformative priors.

$\theta_1 \sim \mathrm{N}(0, 100^2)$, $\theta_3 \sim \mathrm{N}(0, 10^2)$ (both far wider than plausible values), $\theta_2, \theta_4 \sim \mathrm{InvGamma}(0.1, 0.1)$, $\theta_5, \theta_6 \sim \mathrm{Gamma}(0.1, 0.1)$, $\sigma_0^2 \sim \mathrm{InvGamma}(0.1, 0.1)$.

(c) Convergence

(our Gibbs sampler: three chains from very different starting hyperparameters, 25,000 iterations each, 5,000 burn-in):

	Mean	SD	95% interval	$\hat{R}$	ESS (of 60,000)
θ_1 (mean pace)	6.182	0.035	(6.114, 6.251)	1.000	19,253
θ_2 (var. of pace)	0.039	0.012	(0.019, 0.066)	1.000	11,979
θ_3 (mean ρ)	0.990	0.016	(0.958, 1.020)	1.000	22,118
θ_4 (var. of ρ)	0.0087	0.0021	(0.0055, 0.0137)	1.000	24,769
θ_5 (shape)	2.27	0.45	(1.49, 3.27)	1.001	1,534
θ_6 (scale)	0.031	0.007	(0.019, 0.048)	1.001	1,720
σ_0	0.177	0.029	(0.126, 0.239)	1.000	9,981



All $\hat{R} \leq 1.001$, overlapping trace plots from dispersed starts; θ_5, θ_6 mix most slowly (they are strongly correlated: shape and scale trade off), but their ESS still exceeds 1,500.

Sanity check

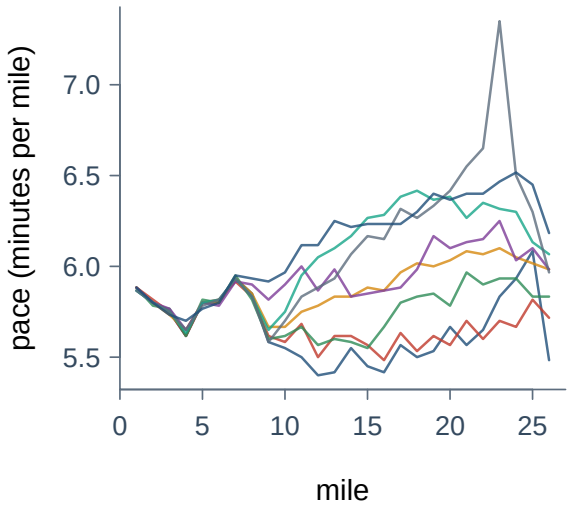
JAGS output (3 chains × 25,000 draws, as the exercise asks).

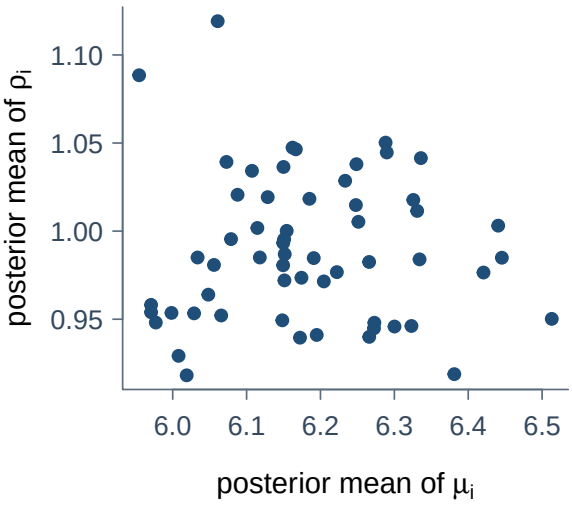
	Mean	95% interval	$\hat{R}$	ESS	our Gibbs mean
θ_1	6.182	(6.114, 6.251)	1.000	25,270	6.182
θ_2	0.0387	(0.019, 0.066)	1.000	11,440	0.0385
θ_3	0.990	(0.958, 1.020)	1.000	29,660	0.990
θ_4	0.0087	(0.0054, 0.0137)	1.000	22,840	0.0087
θ_5	2.26	(1.49, 3.25)	1.001	4,501	2.27
θ_6	0.031	(0.019, 0.047)	1.001	4,640	0.031

All $\hat{R} \leq 1.001$; JAGS and our sampler agree to the third decimal.

› (d) Are the data informative?

Yes, very. θ_1: prior SD 100 → posterior SD 0.035; θ_3: prior SD 10 → 0.016. The inverse-gamma/gamma priors for $\theta_2, \theta_4, \theta_5, \theta_6$ have enormous (or infinite) spread, while the posteriors have 95% widths of 0.046, 0.008, 1.8 and 0.029. The least informed are θ_5, θ_6 (the distribution of 58 variances), which is also why they mix slowest.





› (e) Simplifications and how to compare.

The posterior of θ_3 is ≈ 0.99 and θ_4 is tiny (between-runner SD of ρ_i about 0.09): every runner's pace behaves almost like a random walk with similar persistence. Natural simplifications:

(i) a **common** ρ for all runners ($\theta_4 = 0$);

(ii) a random walk ($\rho_i = 1$), i.e. model the mile-to-mile *changes*;

(iii) a common σ^2.

Compare with WAIC (or LOO-CV) computed from the same one-step-ahead likelihood. For (i): common $\rho = 1.018$ (1.004–1.031) and

Model	lppd	p_W	WAIC
full (runner-specific ρ_i)	963.3	148.9	-1628.8
common ρ	930.8	91.6	$\mathbf{-1678.3}$

The common-ρ model fits almost as well with 57 fewer effective parameters and has the lower WAIC: the simplification is supported. (ρ slightly above 1 means deviations slowly *grow*: runners who slow down tend to keep slowing.)

Answer

(a) DAG above; μ_i = typical pace, ρ_i = persistence, σ_i^2 = variability, θ's = their population mean/variance (and shape/scale). (b) Vague normal priors for θ_1, θ_3; vague inverse-gamma/gamma for the rest. (c) Converged ($\hat{R} \leq 1.001$, ESS $\geq 1{,}500$); JAGS agrees. (d) Yes: posteriors are far more concentrated than the priors, least so for θ_5, θ_6. (e) A common persistence ρ (or a random walk) is a sensible simplification; by WAIC it is better (-1678 vs. -1629).

Takeaway & common mistakes

When a hierarchical variance (θ_4) is estimated to be small, consider removing that level: it simplifies the model and often improves prediction.

Exercise 7.5 — Exponential survival with censoring: the lung data

Exercise 5 (exact text from the book)

Download the lung data from the survival package in R. Let Y_i be the survival time (days) for subject $i \in \{1, \ldots, n\}$. We will make the unrealistic assumption that the survival times follow an exponential distribution, $Y_i|\lambda \sim \text{Exponential}(\lambda)$. The study runs from time 0 to time T, with patient i joining the study at time $T - T_i$ so that they are part of the study for T_i days. The analysis is complicated by censoring; i.e., some patients have not experienced their event at the conclusion of the study. An event is observed if $Y_i < T_i$ and censored if $Y_i > T_i$, with $\delta_i = I(Y_i > T_i)$ denoting the binary indicator that the event is censored. For uncensored patients with $\delta_i = 0$, the contribution to the likelihood function is the exponential PDF, $\lambda \exp(-\lambda y_i)$; for censored patients with $\delta_i = 1$, the contribution to the likelihood function is the probability that they survive beyond T_i days, $\exp(-\lambda T_i)$.

(a) Write the likelihood function in terms of the parameter λ and the data $(Y_1, T_1, \delta_1), \ldots, (Y_n, T_n, \delta_n)$.

(b) Derive a conjugate prior distribution for λ, and propose hyperparameters that lead to an uninformative prior distribution.

(c) Using the full dataset, summarize the posterior distribution of λ under a few conjugate priors (i.e., different hyperparameters). Are the results sensitive to the prior?

(d) Using an uninformative conjugate prior, analyze the data separately for men and women, and test whether the survival distribution varies by sex.

In plain words

Estimate the death rate of 228 advanced lung-cancer patients, many of whom were still alive at the end of follow-up, and compare men and women.

Background you need

Censoring: primer, point 4 (and Exercise 2.12). In R's `lung` data, `time` is the observed follow-up $t_i = \min(Y_i, T_i)$ and `status` is 2 for a death, 1 for censored.

Key idea

The likelihood depends only on the number of deaths $d = \sum_i (1 - \delta_i)$ and the total time at risk $\sum_i t_i$; it is a Gamma kernel in λ.

Solution

(a) Likelihood.

$$L(\lambda) = \prod_{i=1}^{n} \left[\lambda e^{-\lambda Y_i}\right]^{1-\delta_i} \left[e^{-\lambda T_i}\right]^{\delta_i} = \lambda^{\sum_i (1-\delta_i)} \exp\left\{ -\lambda \sum_i \left[(1-\delta_i) Y_i + \delta_i T_i\right] \right\} = \lambda^{d} e^{-\lambda \sum_i t_i}.$$

(b) Conjugate prior.

$L(\lambda)$ has the form $\lambda^{\text{power}} e^{-\lambda \times \text{constant}}$, so a $\text{Gamma}(a, b)$ prior is conjugate:

$$p(\lambda \mid \text{data}) \propto \lambda^{d} e^{-\lambda \sum t_i} \cdot \lambda^{a-1} e^{-b\lambda} \;\Rightarrow\; \lambda \mid \text{data} \sim \text{Gamma}\Big(a + d,\; b + \sum_i t_i\Big).$$

Uninformative: small a and b, e.g. $a = b = 0.01$ (worth 0.01 deaths in 0.01 days of follow-up).

› (c) Full data

($n = 228$, $d = 165$ deaths, 63 censored, $\sum t_i = 69{,}593$ days):

Prior	Posterior	Mean rate/day	median survival	95% interval
$\text{Gamma}(0.01, 0.01)$	$\text{Gamma}(165.01,\ 69{,}593)$	0.00237	292 days	(0.00202, 0.00275)
$\text{Gamma}(0.1, 0.1)$	$\text{Gamma}(165.1,\ 69{,}593)$	0.00237	292 days	(0.00202, 0.00275)
$\text{Gamma}(1, 1)$	$\text{Gamma}(166,\ 69{,}594)$	0.00239	291 days	(0.00204, 0.00276)
$\text{Gamma}(10, 1000)$	$\text{Gamma}(175,\ 70{,}593)$	0.00248	280 days	(0.00213, 0.00286)

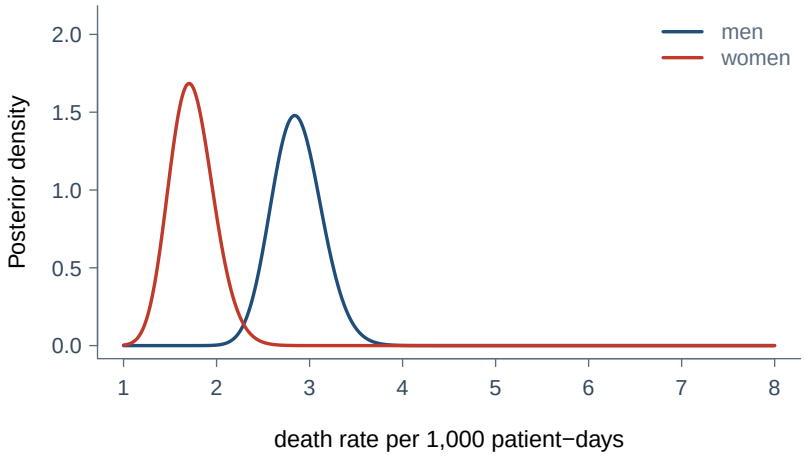
(Median survival $= \log 2/\lambda$.) Not sensitive: vague priors are indistinguishable, and even the informative $\text{Gamma}(10, 1000)$ prior (worth 10 deaths in 1,000 days, mean 0.01/day) moves the rate by only 4%.

› (d) Men vs. women.

Men: 112 deaths in 39,086 days; women: 53 deaths in 30,507 days. With $\text{Gamma}(0.01, 0.01)$ priors:

$$\lambda_{\text{men}} \approx 0.00287, \quad \lambda_{\text{women}} \approx 0.00174 \text{ per day}; \qquad \text{Prob}(\lambda_{\text{men}} > \lambda_{\text{women}} \mid \text{data}) = 0.999,$$

hazard ratio men/women: median 1.65, 95% CI (1.20, 2.31).



A Bayes factor for "separate rates" vs. "common rate" (closed form, $\text{Gamma}(a, b)$ prior on each rate):

Prior on each rate	prior mean rate	BF(separate : common)
$\text{Gamma}(1,\ 0.001)$ (absurdly vague)	1000 per day	0.0001
$\text{Gamma}(1,\ 100)$	0.01 per day	8.0
$\text{Gamma}(1,\ 420)$ (centred on the pooled rate)	0.0024 per day	16.4
$\text{Gamma}(2,\ 840)$	0.0024 per day	22.4

With any *sensible* prior the BF is 8–22 (substantial to strong evidence for different rates); a ridiculously vague prior reverses it (Lindley's paradox again).

Answer

(a) $L(\lambda) = \lambda^d e^{-\lambda \sum_i t_i}$ with $d = \#\text{deaths}$ and t_i the follow-up times. (b) $\lambda \sim \text{Gamma}(a, b) \Rightarrow \text{Gamma}(a + d,\ b + \sum t_i)$; uninformative $a = b = 0.01$. (c) $\lambda \approx 0.0024$/day (median survival ≈ 290 days), insensitive to the prior. (d) Men die at about 1.65 times the rate of women (95% CI 1.20–2.31; Prob $= 0.999$); BF 8–22 under sensible priors: survival differs by sex.

Takeaway & common mistakes

Posterior probabilities of a direction ($\lambda_m > \lambda_f$) are robust; Bayes factors need a prior that is realistic for the scale of the problem.

Exercise 7.6 — A spatial model for village malaria prevalence

Exercise 6 (exact text from the book)

Download the Gambia data

```
> library(geoR)
> data(gambia)
> ?gambia
```

The data consist of 2,035 children that live in 65 villages. For village $v \in \{1, \ldots, 65\}$, denote n_v as the number of children in the sample, Y_v as the number of children that tested positive for malaria, and p_v as the true probability of testing positive for malaria. We use the spatial model $\alpha_v = \text{logit}(p_v)$, where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_{65})^T$ follows a multivariate normal distribution with mean $\text{E}(\alpha_v) = \mu$, variance $\text{Var}(\alpha_v) = \sigma^2$ and correlation $\text{Cor}(\alpha_u, \alpha_v) = \exp(-d_{uv}/\rho)$, where d_{uv} is the distance between villages u and v. For priors assume $\mu \sim \text{Normal}(0, 10^2)$, $\sigma^2 \sim \text{InvGamma}(0.1, 0.1)$, and $\rho \sim \text{Uniform}(0, d^*)$ where d^* is the maximum distance between villages.

(a) Specify the data layer, process layer and prior layer for this hierarchical model.

(b) Fit the model using JAGS and assess convergence.

(c) Summarize the data and results using five maps: the sample size n_v, the sample proportion Y_v/n_v, the posterior means of the p_v, the posterior standard deviations of the p_v, and the posterior probabilities that p_v exceeds 0.5.

In plain words

Model each village's malaria prevalence so that nearby villages tend to be similar, and map the results.

Background you need

Gaussian-process random effects and elliptical slice sampling: primer, point 5.

Key idea

The village logits $\boldsymbol{\alpha}$ are a spatially correlated random field. Collapsing σ^2 out of the update for the range ρ greatly improves mixing (the two are strongly dependent).

Solution

(a) The three layers.

- **Data layer**: $Y_v \mid p_v \overset{indep}{\sim} \text{Binomial}(n_v, p_v)$, $v = 1, \ldots, 65$.
- **Process layer**: $\alpha_v = \text{logit}(p_v)$, $\boldsymbol{\alpha} \sim \text{N}\big(\mu \mathbf{1},\ \sigma^2 \mathbf{R}(\rho)\big)$, $R_{uv} = \exp(-d_{uv}/\rho)$.
- **Prior layer**: $\mu \sim \text{N}(0, 10^2)$, $\sigma^2 \sim \text{InvGamma}(0.1, 0.1)$, $\rho \sim \text{Uniform}(0, d^*)$ with $d^* = 273$ km.

(b) Fitting.

Our sampler, per iteration: elliptical slice sampling for $\boldsymbol{\alpha}$; $\mu \mid \cdot$ normal; ρ by Metropolis on $\text{logit}(\rho/d^*)$ with σ^2 integrated out; then $\sigma^2 \mid \cdot$ inverse-gamma. Three chains (started at $\rho = 5, 40, 120$ km) $\times$ 20,000 kept draws.

	Mean	SD	95% interval	$\hat{R}$	ESS (of 60,000)
μ	-0.15	1.45	$(-2.81,\ 3.54)$	1.001	7,297
σ	2.00	1.00	(0.93, 4.46)	1.002	705
ρ (km)	66	69	(8, 252)	1.006	430

Why? ρ and σ are weakly identified from 65 villages (a long range with a large variance looks much like a short range with a smaller variance), so their posteriors are wide and they mix slowest. The collapsed ρ update raised their ESS from about 65–90 to 430–700.

Village probabilities p_v: $\hat{R}$ between 1.000 and 1.046, ESS 70–several thousand (median 186). This is adequate for posterior means and probabilities in the maps, though a longer run would tighten the Monte Carlo error for a few villages with no positive children.

Sanity check

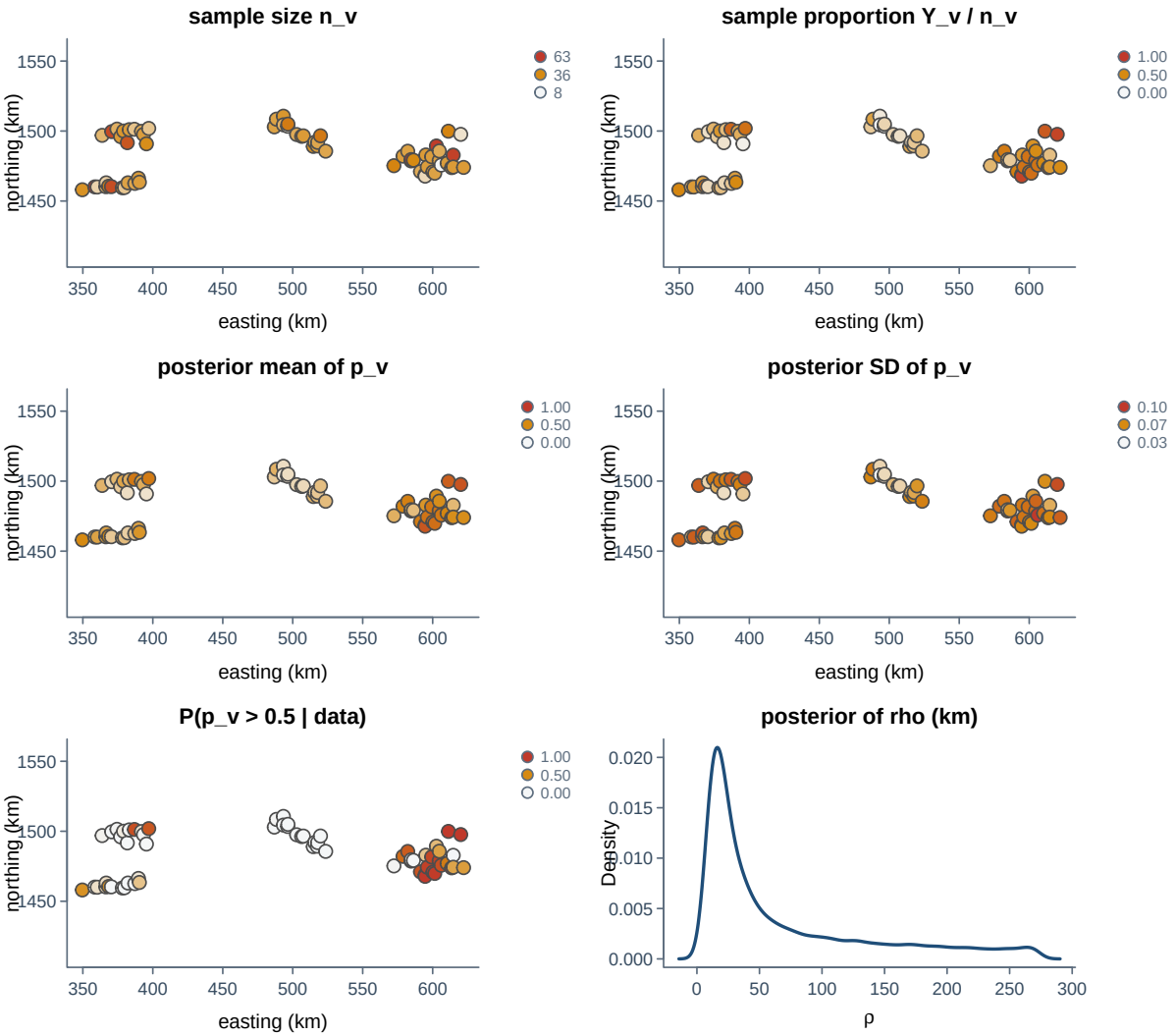
JAGS output (3 chains × 4,000 draws — JAGS inverts a 65×65 matrix at every update, so only a short run is practical).

	Mean	95% interval	$\hat{R}$	ESS
μ	-0.11	$(-3.07,\ 3.84)$	1.007	2,260
σ	2.12	(0.94, 4.50)	1.023	107
ρ (km)	79	(8, 256)	1.022	125

Consistent with our sampler (whose longer, collapsed-update chains mix better): the same wide, weakly identified range and scale.

(c) Five maps

(plus the posterior of ρ):



- **Sample sizes** range from 8 to 63 children per village.
- **Sample proportions** are noisy, from 0 to 0.93; the eastern cluster has the highest rates.
- **Posterior means** are smoothed versions of the sample proportions (correlation 0.98): villages with few children are pulled toward their neighbours.
- **Posterior SDs** are largest for small villages and for villages whose rate differs from their neighbours'.
- $\text{Prob}(p_v > 0.5 \mid \textbf{data})$: 10 villages exceed 0.9 — all of them in the eastern cluster (easting 595–620 km) — natural targets for intervention.

Answer

(a) Data: $Y_v \sim \text{Bin}(n_v, p_v)$; process: $\text{logit}(\mathbf{p}) \sim \text{N}(\mu\mathbf{1}, \sigma^2 e^{-\mathbf{D}/\rho})$; priors as given. (b) Converged ($\hat{R} \leq 1.006$ for μ, σ, ρ; JAGS agrees); $\sigma \approx 2.0$, range $\rho \approx 66$ km but very uncertain (8–252 km). (c) The maps show a high-prevalence eastern region; 10 villages have $\text{Prob}(p_v > 0.5) > 0.9$.

Takeaway & common mistakes

Spatial models turn noisy village-level counts into a smooth risk map with honest uncertainty — exactly what is needed to target malaria control.